

FAST – NUCES



Object Oriented Programming Lab Manual

Course Teacher	Ms. Hina Iqbal
Lab Instructor	Mr. Ahmad Jawad Mustasim
Lab TA	Ms. Mania Shakeel
Section	BSE-2B
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Department of Computer Science

FAST-NU, Lahore, Pakistan

Lab 11 – Inheritance in OOP

Objectives

- Single Inheritance
 - Multi-level Inheritance
 - Hierarchical Inheritance
 - Multiple Inheritance
 - Method overriding
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Read carefully and attempt following task

Become a Pokémon Master

You're stepping into the world of Pokémon as a developer! Your mission is to design a mini Pokémon system using Object-Oriented Programming concepts in C++.



Welcome to the world of Pokémon!

Instructions

- You are **allowed to use built-in string** in this lab

Part 1: Multi-Level Inheritance

1. Base Class: Trainer

Attributes:

- name (string)
- age (int)

Functions:

- setName(string name)
- setAge(int age)
- showInfo()

2. Derived Class: GymTrainer (inherits from Trainer)

Attributes:

- gymName (string)
- specialtyType (string)

Functions:

- setGym(string)
- setType(string)
- showInfo()

3. Derived Class: GymLeader (inherits from GymTrainer)

Attributes:

- badgeReward (string)

Functions:

- setBadge(string)
- showInfo()

Part 2: Multiple Inheritance

4. Class: Professor

Attributes:

- field (string)

Functions:

- setField(string)
- showProfessorInfo()

5. Class: ProfessorTrainer

(Inherits from **Trainer** and **Professor**)

Attributes:

- researchArea (string)

Functions:

- setResearch(string)
- showInfo()

Part 3: Hierarchical Inheritance

6. Base Class: Pokemon

Attributes:

- name (string)
- level (int)

Functions:

- setPokemon(string, int)
- showInfo()

Pokémon Types (Derived Classes)

You must implement:

7. FirePokemon

- flamePower (int)

8. WaterPokemon

- waterPressure (int)

9. ElectricPokemon

- voltage (int)

10. GrassPokemon

- growthRate (int)

11. FlyingPokemon

- flightSpeed (int)

Observation Task

In main():

- Create:
 - A simple Pokemon object

- Objects of all derived Pokémon types
- Call showInfo() for each

Observe carefully:

- How the **same function name behaves differently**
- How derived classes **override** base class behavior

Part 4: Independent Class

12. Class: Badge (Standalone Class)

This class does **NOT inherit** from any class.

Attributes:

- badgeID (int)
- badgeInfo (string)

Functions:

- setBadge(int, string)
- showBadge()

Purpose:

To show that **not all classes must participate in inheritance**

Main Function Requirements

Create objects of:

- Trainer
- GymTrainer
- GymLeader
- ProfessorTrainer
- Badge
- All Pokémon types

Call:

- All setter functions
- All showInfo() functions

Object examples:**Trainers**

- Ash Ketchum, age 15
- Misty, age 16
- Gary Oak, age 16

Gym Trainers

Trainer working under gym:

- GymTrainer 1
 - Name: *Daisy*
 - Age: 18
 - Gym: *Cerulean Gym*
 - Type: *Water*
- GymTrainer 2
 - Name: *Flint*
 - Age: 22
 - Gym: *Cinnabar Gym*
 - Type: *Fire*

Gym Leaders

- Brock
 - Age: 20
 - Gym: *Pewter Gym*
 - Type: *Rock*
 - Badge: *Boulder Badge*
- Lt. Surge
 - Age: 25
 - Gym: *Vermilion Gym*
 - Type: *Electric*
 - Badge: *Thunder Badge*

Professors

- Professor Oak, field: Pokémon Evolution

ProfessorTrainer

- Professor Willow
 - Age: 35
 - Field: Pokémon Behavior
 - Research Area: Field Research

Pokémon

- Pikachu → Electric (voltage: 90)
- Charizard → Fire (flamePower: 120)
- Squirtle → Water (waterPressure: 80)
- Bulbasaur → Grass (growthRate: 70)
- Pidgeotto → Flying (flightSpeed: 85)

Also create:

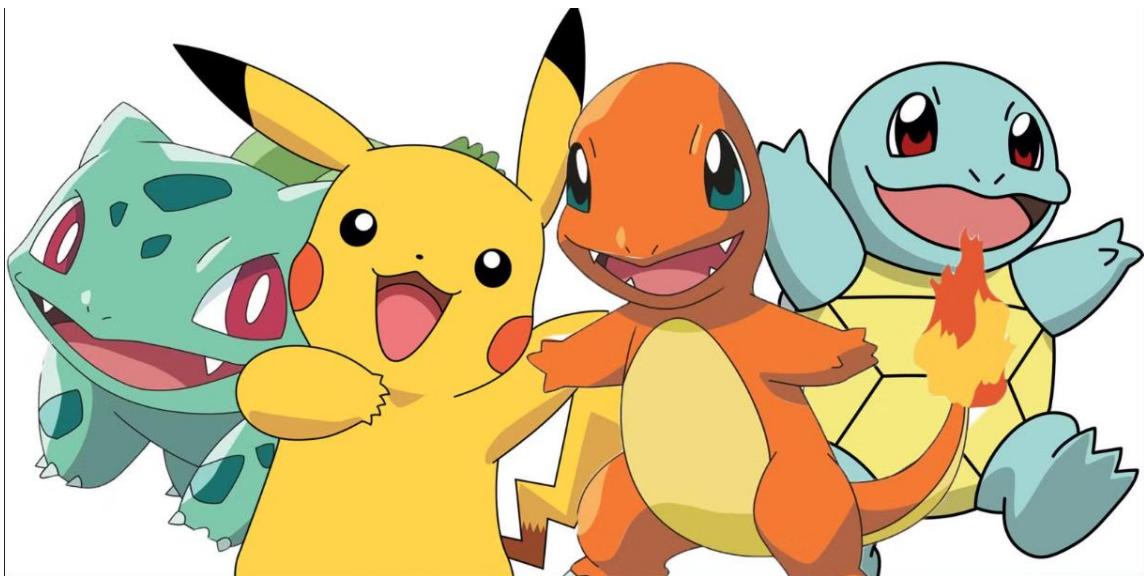
- Eevee - One simple Pokemon object (base class only)

Badges

- ID: 101 → Thunder Badge
- ID: 102 → Cascade Badge
- ID: 103 → Boulder Badge

TO BE CONTINUED IN NEXT LABS.....

-----Happy Coding! 😊-----



-----Submission Instructions-----

- Submit your tasks on Google Classroom within given deadline
 - Submit one cpp file only.
 - Also submit screenshots of your task
 - “.cpp” is the actual file where all your code is saved.
 - Always submit .cpp files when asked for submission
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